New Phytologist

***New Phytologist* Supporting Information**

Article title: PEP725: fifteen years of driving European and global phenology science

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The following Supporting Information is available for this article:

Fig. S1 Number of new user registrations on the PEP725 website. Data for 2015 are unavailable due to a system failure.

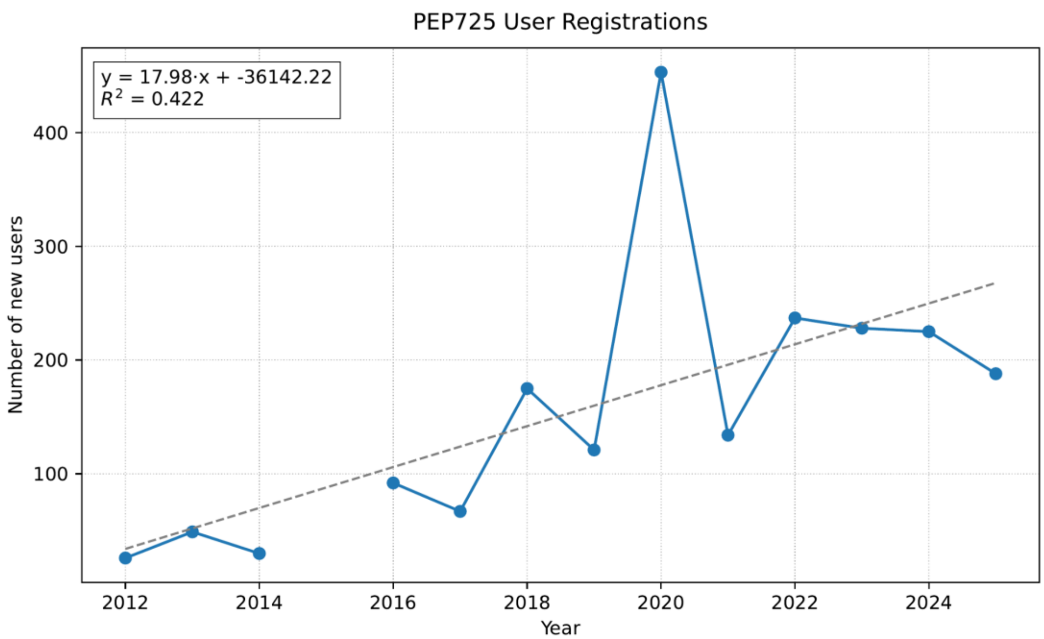

Fig. S2 Relative distribution of countries of origin among PEP725 users. The figure presents the average user distribution for the years 2024 and 2025. China shows by far the highest demand for PEP725 phenological data, representing more than 26% of all users, followed by Spain (11%), Austria (9%) and Germany (8%). At the continental basis, nearly two-thirds of users originate from Europe, while about 30% from Asia. Because inactive accounts are removed after two years, long-term user statistics cannot be derived.

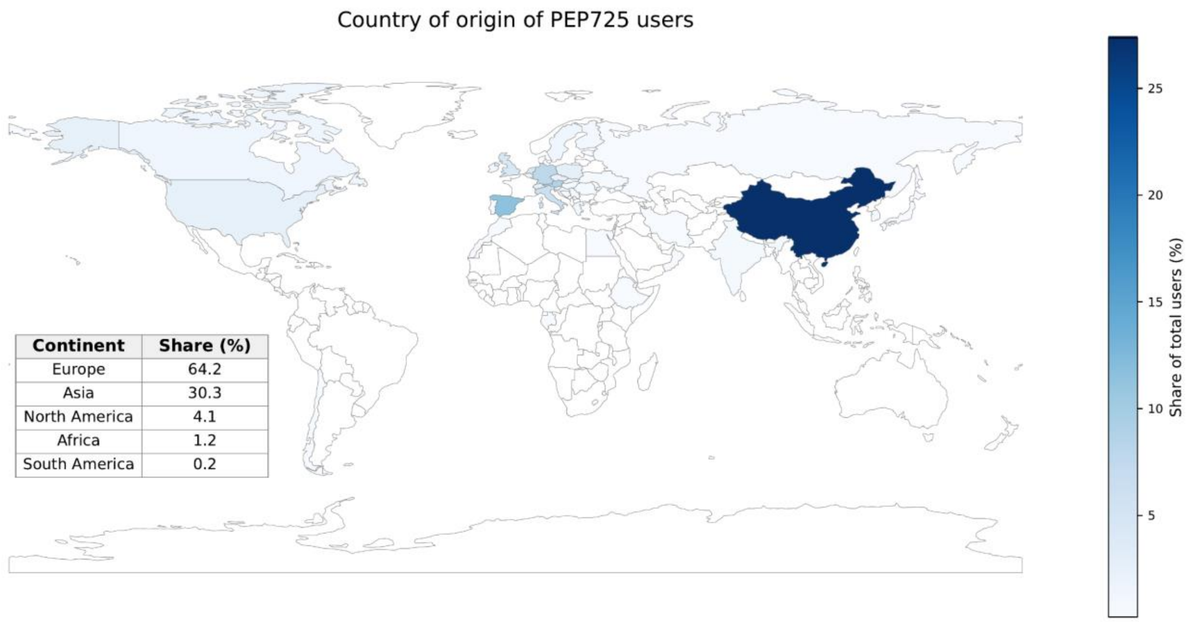

Table S1 List of acronyms.

Acronym	Full term / name	Explanation or context of use
API	Application Programming Interface	Standardized interface enabling automated data access to PEP725 and other web services.
BBCH	Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie scale	Widely used phenological coding system describing plant growth stages (Meier, 2001).
COST-725	European Cooperation in Science and Technology Action 725	Precursor collection of phenological data (2005–2009) that harmonized phenological data across Europe and led to the establishment of PEP725.
EBV	Essential Biodiversity Variable	Concept introduced under GEO-BON (Group on Earth Observations Biodiversity Observation Network) for harmonized global biodiversity monitoring; includes phenology as a variable.

EGU	European Geosciences Union	The European Geosciences Union (EGU) is the leading organisation for Earth, planetary and space science research in Europe.
EU	European Union	Political and funding framework supporting European research infrastructures such as PEP725.
ET	EUMETNET Expert Team on Phenology	Technical group coordinating phenological observation methods and metadata standards within EUMETNET.
EUMETNET	European Meteorological Network	Association of European National Meteorological and Hydrological Services (NMHSs) fostering collaboration and data exchange.
EVI	Enhanced Vegetation Index	Remote-sensing metric used to quantify canopy greenness and seasonal dynamics.
FACE	Free-Air CO2 Enrichment	Free-Air CO2 Enrichment, a scientific method for studying the effects of elevated carbon dioxide on plants and ecosystems
FAIR	Findable, Accessible, Interoperable and Reusable	Guiding principles for data management and sharing adopted by PEP725.
FISE	Forest Information System for Europe	The Forest Information System for Europe (FISE), is a single entry point for data and information on forests in Europe. FISE brings together data, information and knowledge gathered or derived through key forest-related policy drivers.
GAPON	Global Alliance of Phenological Observation Networks	Global phenological observation network initiative by the International Society of Biometeorology Phenology Commission (ISB-PC) and the World Meteorological Organization Commission for Agricultural Meteorology (WMO-CAgM)

GCOS	Global Climate Observing System	WMO-sponsored programme defining essential climate variables, including phenology-related terrestrial indicators.
GeoSphere	GeoSphere Austria	Austrian national geological, geophysical, climatological and meteorological service hosting the PEP725 data infrastructure.
GP	Ground Phenology	Ground based phenological observations (as opposed to satellite derived Land Surface Phenology, LSP)
GPS	Global Positioning System	Satellite-based positioning system used for georeferencing observation sites.
HR-VPP	High-Resolution Vegetation Phenology and Productivity	Copernicus Land Service product providing satellite-based phenology metrics for Europe.
IPCC	Intergovernmental Panel on Climate Change	United Nations body assessing climate-change science; PEP725 data contribute to AR6 analyses of phenological responses.
IPG	International Phenological Gardens	Network of cloned indicator trees used for long-term phenology monitoring in Europe.
LSP	Land Surface Phenology	Seasonal dynamics of vegetation derived from satellite observations.
ML	Machine Learning	Multivariate statistical method
MODIS	Moderate Resolution Imaging Spectroradiometer	NASA satellite sensor providing global vegetation indices and phenology products.

NDVI	Normalized Difference Vegetation Index	Remote-sensing index quantifying greenness, used to estimate phenological transitions.
NMHS	National Meteorological and Hydrological Service	National agencies providing meteorological observations integrated in EUMETNET and PEP725.
PEP725	Pan European Phenology Database project 725	European data portal providing harmonized in-situ phenological observations from > 25 national networks.
PMIP	Paleoclimate Modelling Intercomparison Project	International modelling framework; used here in reference to long-term climate–phenology context.
PRISMA-EcoEvo	Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Ecology and Evolutionary Biology	The PRISMA for ecology and evolutionary biology (EcoEvo) provides a 27-item checklist and guidance for reporting systematic reviews and meta-analyses of primary research in ecology and evolutionary biology.
PPO	Plant Phenology Ontology	Semantic vocabulary for structuring and exchanging phenological data.
QC	Quality Control	An assembly of various methods to evaluate the plausibility of observations
PRI (sPRIref)	(Scaled) Photochemical Reflectance Index	The Photochemical Reflectance Index (PRI) is a reflectance measurement, which is sensitive to changes in carotenoid pigments (e.g. xanthophyll pigments) in live foliage.
RCP	Representative Concentration Pathway	Climate-forcing scenario used in model simulations of future phenology.
SIF	Solar-Induced Fluorescence	SIF vegetation is a red light emitted by plants as a byproduct of photosynthesis. By measuring SIF from space, a plant's photosynthetic activity can be monitored remotely.

SOS / EOS	Start of Season / End of Season	Key transition dates in plant phenology derived from observations or remote sensing.
SSP	Shared Socio-economic Pathway	Updated scenario framework extending the RCP concept for future climate projections.
TRY	Plant Trait Database	TRY is a network of vegetation scientists headed by Future Earth and the Max Planck Institute for Biogeochemistry, providing a global database of curated plant traits.
USA-NPN	USA National Phenology Network	American open-data phenology network aligned with FAIR principles.
VI	Vegetation Index	A vegetation index (VI) is a numerical value calculated from a satellite's spectral imaging data to monitor vegetation health and density
WMO	World Meteorological Organization	UN-specialized agency coordinating international meteorological and climatological activities.
ZAMG	Zentralanstalt für Meteorologie und Geodynamik	Former Austrian meteorological service (now part of GeoSphere Austria); initiator and host of the PEP725 project.

Table S2 Key characteristics of the contributing institutions to the PEP725 database. If an institution is listed as "COST-725," the corresponding dataset originates from the COST-725 action and has not been updated since that project ended. This may be due to different circumstances—for example, some national networks discontinued their phenological monitoring (e.g., Belgium), while in other cases data collection was continued or taken over by a different institution (e.g., Poland). Further details on the COST-725 project are provided in Nekovář et al. (2008).

Number	Country	Institution	Begin year	End year	Number of observations	Number of Stations

1	Austria	National Meteorological Service	1775	2025	734487	1573
2	Belgium	COST725	1949	2004	1614	54
3	Bosnia and Herzegovina	National Hydrometeorological Service	1971	2024	9516	6
4	Croatia	National Hydrometeorological Service	1961	2024	40251	11
5	Czech Republic	COST725	1951	2011	15815	135
6	Czech Republic	National Hydrometeorological Service	1961	2024	8028	12
7	Finland	Natural Resources Institute Finland (Luke)	1997	2017	6752	35
8	France	COST725	1963	2005	152	6
9	France	ODS Tela Botanica	2006	2025	22977	1488
10	France	Phénoclim CREA Mont-Blanc	2004	2025	50606	3506
11	France	AgroClim Pheno	1875	1977	6658	3
12	France	Forêt	1872	2024	129299	1058
13	Germany	National Meteorological Service	1951	2024	14098378	6788
14	Germany	School	2014	2018	79	9
15	International	IPG International Phenological Gardens	1962	2009	23492	79

16	Ireland	National Meteorological Service	1966	2024	2716	3
17	Italy	CRI Research and Innovation Centre	1984	2011	181	8
18	Italy	CRA-CMA Agricultural research and experimental organisation	2006	2013	450	62
19	Latvia	COST725	1971	2000	621	8
20	Latvia	University of Latvia	1970	2018	31085	103
21	Lithuania	COST725	1960	2004	27892	30
22	Luxembourg	COST725	1966	2004	183	1
23	Montenegro	National Hydrometeorological Service	1951	2021	13721	7
24	Netherlands	COST725	1868	1968	14610	537
25	North Macedonia	National Hydrometeorological Service	1961	2018	1958	5
26	Norway	COST725	1927	2005	878	4
27	Poland	COST725	1951	1992	6941	13
28	Poland	National Hydrometeorological Service	2007	2018	2682	19
29	Romania	COST725	1982	2005	4220	80

30	Slovakia	National Hydrometeorological Service	2000	2024	109788	76
31	Spain	COST725	1946	2001	1466	15
32	Spain	National Meteorological Service	2013	2020	4696	54
33	Spain	National Meteorological Service	1986	2025	7918	12
34	Sweden	Swedish Agricultural University	2001	2020	2224	5
35	Sweden	National Meteorological Service	1961	2020	35309	14
36	Sweden	COST725	1870	1951	274126	698
37	Sweden	The Swedish National Phenology Network (SLU and SMHI)	1980	2024	134259	1461
38	Switzerland	National Meteorological Service	1951	2024	268412	182
39	UK	Nature's Calendar / Woodland Trust	1950	2005	163440	10509
40	Ukraine	M.M. Gryshko National Botanical Garden NAS of Ukraine	2015	2022	238	1
41	Ukraine	Arboretum Vysokogirny, State Enterprise Nadvirna forestry	2007	2022	96	1

Table S3 Peer-reviewed studies that used data from the PEP725 database (up to 2025).

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Methods S1 Workflow for the systematic identification of scientific literature

To quantify and review the scientific impact and use of PEP725, we conducted a systematic literature search in *Google Scholar* (https://scholar.google.com/ , accessed in May 2025) using the keyword "PEP725 phenology" in the title, abstract, author keywords, or funding acknowledgment fields. All retrieved records were manually checked to verify that PEP725 data were used in the study; duplicates and false positives (e.g., unrelated uses of the abbreviation "PEP") were removed. The resulting dataset comprised approximately 140 peer-reviewed publication**s** that explicitly used PEP725 data.

For comparison, the foundational PEP725 description paper (*Templ et al., 2018;* DOI: 10.1007/s00484-018-1512-8) has been cited ~235 times according to Google Scholar, but not all of these citations involve direct use of the database, as many refer to it for contextual or methodological background. Thus, the count of 140 papers should be regarded as a conservative estimate of actual data use. Not all authors cite PEP725 consistently or report publications to the database administrators as encouraged on the PEP725 website, and some may use the data without explicit acknowledgment.

These publications were categorized by research domain (e.g., phenological modelling, remote sensing, ecosystem studies) and form the basis of the examples summarized in Sections 2 and 3.

Notes S1 Overview of the funding sources and measures ensuring the sustainability of the PEP725 database.

The initial establishment of PEP725 was made possible through the COST Action 725 (2004–2009; https://www.cost.eu/actions/725/, titled: Establishing a European Phenological Data Platform for Climatological Applications), which provided the first European framework for coordinating national phenological networks. Since the end of the COST Action, operational continuity has been guaranteed by GeoSphere Austria. Data management, software development and coordinating activities have been supported by EUMETNET (The European Meteorological Network; https://eumetnet.eu/), which ensures institutional embedding within the meteorological community. In-kind contributions from GeoSphere Austria supplement the EUMETNET funding. Apart from these core supports, the individual PEP725 members and contributors rely solely on their own funding for the operation of their networks. Establishing a stable, multi-annual funding structure, for example through European research-infrastructure or environmental-monitoring frameworks would secure the sustainability and further development of this unique transnational resource.

Notes S2 Citation of PEP725-based studies in the IPCC report

Phenology is referenced 226 times in *Climate Change 2022: impacts, Adaptation and Vulnerability* (IPCC AR6, Working Group II). It plays a central role in the assessment, particularly in discussions of climate-change impacts across ecosystems.
Below we provide the list of IPCC WGII chapters that cite studies based on PEP725 data:

2.4.2.4 Observed Phenological Responses to Climate Change: Piao et al. (2019), Menzel et al. (2020)
2.4.2.5. Observed Complex Phenological and Range Shift Responses: Ettinger et al. (2020)
Table 2.2 Menzel et al. (2020)
Box 5.2: Case Study: Wine: Martinez-Lüscher et al (2016)
Chapter 13.3.1.2 Projected Risks for Terrestrial and Freshwater Ecosystems: Wu et al (2018)
Wang H et al (2020): erroneously this work has been cited in the IPCC text, but is not in the references section.

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